Experiment 9.1

Simulation of Ground Wave and Sky Wave Propagation using MATLAB

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Aim

To simulate ground wave and sky wave propagation and analyse the field strength, maximum usable frequency (MUF) and skip distance.

Software Used

MATLAB

Theory / Formulas

Ground wave field strength: $E_{ground} = (300 \sqrt{P_t}) / d$ (mV/m), where P_t is the transmitted power (kW) and d the distance (km).
Sky wave propagation depends on the MUF and the skip distance, the minimum distance at which the sky wave returns to earth.

Procedure

1. Open MATLAB and create a new script.
2. Define the parameters and generate the signals/data.
3. Compute the ground wave field strength versus distance.
4. Compute the MUF and the skip distance.
5. Plot the individual and combined propagation models.

Output

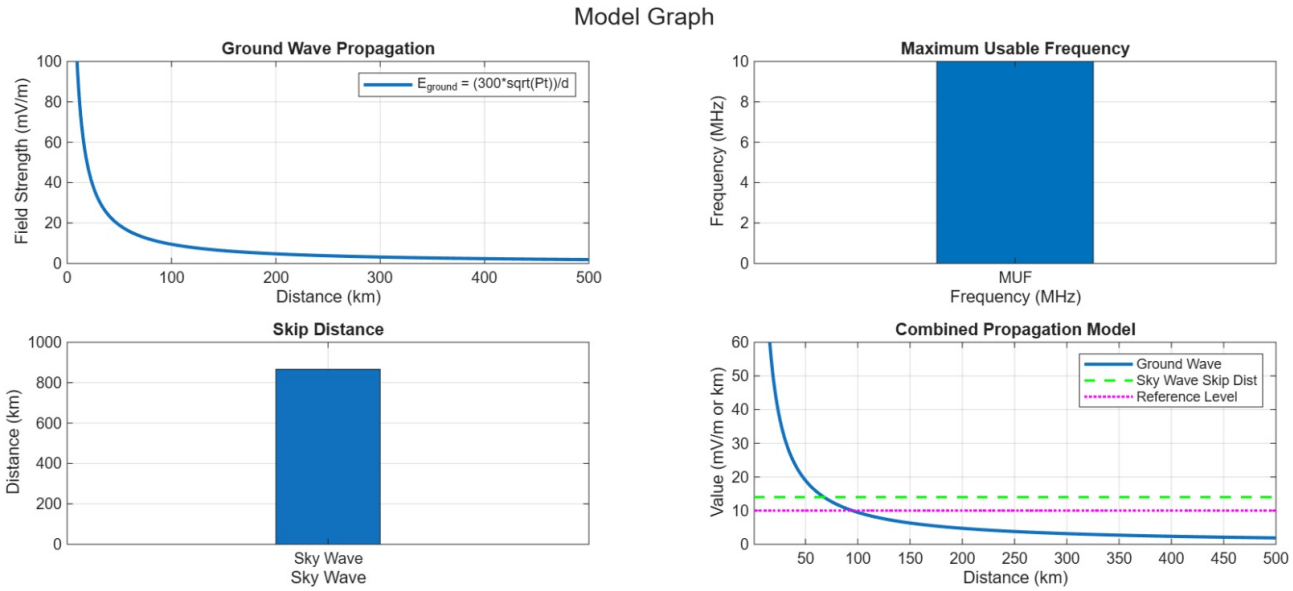


Fig. 1: Ground wave propagation, maximum usable frequency, skip distance and combined propagation model

Result

The ground wave field strength decreases with distance, while the sky wave covers distances beyond the skip distance. The combined model compares both propagation modes.